

Fis Net 09.02.2023



2] Sistema di bosoni

100 bosoni non interagenti privi di spin
 $E_n = \epsilon n$ ($n \in \mathbb{N}$)

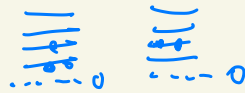
• Quanti e quali bosoni con $E = 4\epsilon$

$$4 = 1 + 1 + 1 + 1$$

$$= 1 + 1 + 2$$

$$= 1 + 3$$

$$= 4$$



$$\langle M_0 \rangle = \frac{1}{e^{-\beta\mu} - 1} = 99 \quad \lambda = \emptyset \quad E = \emptyset$$

$$\langle M_1 \rangle = \frac{1}{e^{\beta(\epsilon - \mu)} - 1} = 1$$

we have

$$-\beta\mu = \ln(100/99) = 1.005 \times 10^{-2}$$

$$\beta(E-\mu) = \ln z = 0.693$$

Hence

$$\beta = 0.692/E ; \quad \mu = -0.01445 E$$

NEGATIVE

1] Ising model statistical N particles

$$\begin{array}{c} \xrightarrow{+E} \\ \xrightarrow{-E} \end{array}$$

MICROCANONICAL

$$N = M_+ + M_-$$

Before:

$$E = M_+ E - M_- E$$

$$r = \frac{E}{NE}$$

$$M_+ = N(1+r)/2$$

$$M_- = N(1-r)/2$$

$$\Omega(E, N) = \frac{N!}{M_+! M_-!}$$

$$S = k \ln \Omega$$



using $N! \approx N \ln N - N$ we get

$$\lim_{N \rightarrow \infty} \frac{S}{N} = k \left[\ln 2 - \frac{1}{2} (1+x) \ln(1+x) - \frac{1}{2} (1-x) \ln(1-x) \right]$$

CANONICAL

$$Z = (e^{-\beta \epsilon} + e^{\beta \epsilon})^N$$

$$F = U - TS$$

$$U = \langle E \rangle = - \left(\frac{\partial \ln Z}{\partial \beta} \right)_N = -N \epsilon \tanh(\beta \epsilon)$$

define;

$$x = \frac{U}{N \epsilon} = -\tanh(\beta \epsilon)$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$$

Obtain T ;

$$\beta \epsilon = \frac{1}{2} \ln \left(\frac{1-x}{1+x} \right) = \frac{1}{2} [\ln(1-x) - \ln(1+x)]$$

;

$$U = N \epsilon x$$

$$F = -kT \ln Z = -NkT \left[\ln 2 - \frac{1}{2} \ln(1+x) - \frac{1}{2} \ln(1-x) \right]$$

$$S = \frac{U - F}{T} = Nk \left[\ln 2 - \frac{1}{2} (1+x) \ln(1+x) - \frac{1}{2} (1-x) \ln(1-x) \right]$$

$$\frac{S}{N} = k \left[\dots \right]$$

GRAND CANONICAL

$$\hat{Z} = \sum_{N=0}^{\infty} e^{\beta \mu N} Z(T, N) = \frac{1}{1 - e^{\beta \mu} (e^{\beta \epsilon} + e^{-\beta \epsilon})}$$

$$\Omega = kT \ln \hat{Z}$$

$$\Omega = TS - U + \mu N$$

$$N = e^{\beta \mu} \left(\frac{\partial \ln \hat{Z}}{\partial e^{\beta \mu}} \right) = \frac{e^{\beta \mu} (e^{\beta \epsilon} + e^{-\beta \epsilon})}{1 - e^{\beta \mu} (e^{\beta \epsilon} + e^{-\beta \epsilon})}$$

or standard formula (identical)

$$\langle E \rangle = - \frac{\partial \ln \hat{Z}}{\partial \beta} = - \epsilon \frac{e^{\beta \mu} (e^{\beta \epsilon} - e^{-\beta \epsilon})}{1 - e^{\beta \mu} (e^{\beta \epsilon} + e^{-\beta \epsilon})} = -N \epsilon \tanh(\beta \epsilon)$$

$$\mathcal{H} = \frac{E}{N\epsilon} = -\tanh(\beta\epsilon)$$

$$\beta\epsilon = \frac{1}{2} [\ln(1-x) - \ln(1+x)]$$

$$\beta\mu = \ln\left(\frac{N}{N+1}\right) - \ln 2 + \frac{1}{2} [\ln(1+x) + \ln(1-x)]$$

We obtain

$$E = N\epsilon x$$

$$\mathcal{Q} = kT \ln \Omega = kT \ln(N+1)$$

$$S = \frac{1}{T} (\mathcal{Q} + U - \mu N) =$$

$$= kN \left[\frac{\ln(N+1)}{N} + \ln\left(1 + \frac{1}{N}\right) \right] + kN \left[\ln 2 - \frac{1}{2} (1+x) \ln(1+x) - \frac{1}{2} (1-x) \ln(1-x) \right]$$

$$\frac{S}{N} = k \left[\ln 2 - \frac{1}{2} (1+x) \ln(1+x) - \frac{1}{2} (1-x) \ln(1-x) \right]$$

$\ln N \rightarrow 0$

